

Cloud + DevOps

Master Roadmap

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COMMITMENT	DURATION	CERT BUDGET	UPDATED
50 hrs / week	9 months	~Rs. 64,100	02 Oct 2026

Goal: Remote USD DevOps/Cloud role · **Secondary:** WSO2, Sysco LABS, IFS, Virtusa

Single source of truth. Prices verified Sep 2026 (1 USD ≈ 328 LKR).

Current Status

Phase 1: Foundations (in progress)

Linux ✓ → Bandit 0-20 ✓ → Bash (shellscript.sh) ✓ → Addressbook CLI ✓ → **NEXT:**
Backup script (hero project seed) → System monitor (Python) → CS50P → Dotfiles →
Git/YAML/Networking gaps

Core Principles

If a phase conflicts with these, the principle wins.

- 1. One growing hero project, not six tutorial projects.** Every phase extends the previous phase's work.
- 2. Build first, then certify.** Finish the phase project before booking the phase exam.
- 3. 5 services, not 500.** EC2, S3, VPC, IAM, Lambda. Master these deeply before anything else.
- 4. Concepts over syntax.** Declarative infra, idempotency, immutability, orchestration outlast tool versions.
- 5. Public work beats private brilliance.** Weekly commits, one blog post per phase.

The Hero Project Chain

One idea, extended every phase: a **self-hosted backup + monitoring service**.

Phase	What you add	Result
1 Foundations	Bash script: backs up a folder, timestamps, logs, rotates old files	Runs via cron on Ubuntu. Public on GitHub
2 AWS Core	Rewrite in Python, deploy as Lambda, S3 with lifecycle rules, IAM role, CloudWatch logs	Serverless on free tier, ~\$0/month
3 Containers	Flask/FastAPI companion API (list/restore backups). Docker Compose locally, k3s in cloud	Multi-container app on k3s (Oracle Cloud free tier)
4 CI/CD	GitHub Actions: test, build, Trivy scan, push image, deploy to k3s	Push code, it deploys. Zero manual steps
5 IaC	Terraform provisions AWS infra, Ansible configures k3s node, remote state	terraform destroy + apply rebuilds everything in <10 min
6 Observability	Prometheus + Grafana dashboards, Alertmanager to email	Production-grade CV project with architecture diagram

PHASE 1

Foundations

Months 1-2

Linux, Bash, Git, Python, YAML, Networking

Skills

- ✓ Linux fundamentals: Linux Journey (Grasshopper + Journeyman)
- ✓ OverTheWire Bandit levels 0-20
- ✓ Bash scripting: shellscrip.sh curriculum complete
- ✓ Addressbook exercise (menu-driven, colon-delimited flat file) (*done 02 Oct 2026*)
- ☐ *Optional:* Dave Eddy "Complete Bash Scripting Course" (YouTube, ysap.sh). Watch only: quoting/word splitting, traps, arrays, process substitution, shellcheck. ~4-6 hrs
- ☐ Git & GitHub: branches, PRs, rebase basics, commit hygiene, SSH auth (already using daily; close gaps)
- ☐ Python: CS50P (Harvard) → certificate
- ☐ YAML & JSON: indentation, anchors, lists vs maps
- ☐ Networking: TCP/IP, DNS, HTTP/S, ports, CIDR (compress, CE degree covers most)

KodeKloud Linux Basics is now **paid**. Skipped. Linux Journey + Bandit already cover it.

Resources

Resource	Type	Cost
Linux Journey	Tutorial	Free ✓
OverTheWire Bandit	Hands-on	Free ✓
shellscript.sh	Tutorial	Free ✓
Dave Eddy Bash Course (ysap.sh)	YouTube	Free (optional)
ShellCheck (shellcheck.net)	Linter	Free
Harvard CS50P	Course + cert	Free
MIT Missing Semester	Video course	Free
Learn Git Branching	Interactive	Free
GitHub Foundations (GH-900)	Cert	Free via GitHub Student Pack

Projects (in order)

- ✓ Addressbook CLI (Bash)
- ☐ **Backup script (Bash)** ← **NEXT**. Hero project seed. Lives in `bash/projects/backup-script/`
 - ▶ Source folder as arg/config
 - ▶ Timestamped archive: `tar + $(date +%Y%m%d_%H%M%S)`
 - ▶ Log every run (what, when, success/fail)
 - ▶ Rotate backups older than N days
 - ▶ Error handling: missing source, disk full, permission denied, `set -euo pipefail`, `trap`
 - ▶ Headless, runs from `cron`
 - ▶ Passes `shellcheck` clean
- ☐ System monitor (Python): CPU/RAM check every 5 min, email on threshold, runs as systemd service (*after CS50P*)
- ☐ Dotfiles repo: `.bashrc`, `.vimrc`, tmux config in Git

Milestone check

- ☐ Bash script safely handles filenames with spaces; correct `while IFS= read -r` usage
- ☐ Can explain `ls | grep foo` at process/file-descriptor level
- ☐ Public GitHub with 3+ commits/week for the last month

PHASE 2

AWS Cloud Core

Months 3-4

5 core services deeply, not 500 shallowly

Skills

- ☐ EC2: instance types, key pairs, security groups, EBS, AMIs
- ☐ S3: buckets, lifecycle rules, versioning, presigned URLs, static hosting
- ☐ VPC: subnets, route tables, NAT, SG vs NACL, VPC endpoints
- ☐ IAM: users, roles, policies, least privilege, assume-role, MFA
- ☐ Lambda + API Gateway: triggers, IAM roles, CloudWatch logs
- ☐ CloudWatch: metrics, log groups, alarms
- ☐ Python for AWS: `boto3`, `argparse`, `os`, `subprocess`, `json`, `yaml`, `requests`

Resources

Resource	Type	Cost
AWS Skill Builder (free tier)	Official courses	Free
AWS Educate	Labs + credits + 50% voucher	Free (uni email)
AWS Cloud Quest: Cloud Practitioner	Gamified labs	Free
Andrew Brown CLF-C02 (freeCodeCamp)	14-hr video	Free
Stephane Maarek CLF-C02 (Udemy)	Paid course	~\$15 on sale only
AWS Free Tier	Real usage	Free

Projects

- ☐ Static portfolio: S3 + CloudFront + Route 53 custom domain (warmup)
- ☐ **Hero**: backup service as Python Lambda, EventBridge cron, S3 with lifecycle rules
- ☐ *Optional*: serverless URL shortener (Lambda + API Gateway + DynamoDB)

Certification: AWS Cloud Practitioner (CLF-C02)

- ▶ Cost: \$100 (~Rs. 32,800). With AWS Educate 50% voucher: **\$50 (~Rs. 16,400)**
 - ▶ 65 questions, 90 min, 700/1000 to pass
 - ▶ Passing unlocks 50% off your next AWS exam
 - ▶ Note: AWS Educate 100% free vouchers ended 31 Aug 2025
- ☐ Pass exam

The 20% of Docker/K8s that covers 80% of real use

Skills

- ☐ Docker: images, layers, Dockerfile, ENTRYPOINT vs CMD, volumes, networks
- ☐ Docker Compose: multi-container, healthchecks, depends_on
- ☐ Kubernetes: Pods, Deployments, Services, ConfigMaps, Secrets, Ingress
- ☐ kubectl fluency: write manifests without Googling basics
- ☐ Registries: Docker Hub, GHCR, AWS ECR

Resources

Resource	Type	Cost
Docker official getting-started	Docs	Free
KodeKloud Docker / K8s for Beginners	Course + badge	Verify still free before relying
Play with Docker	Browser labs	Free
Killercoda	K8s browser labs	Free
Kubernetes.io tutorials	Docs	Free
TechWorld with Nana K8s crash course	YouTube	Free

Projects

- ☐ **Hero:** Dockerise backup code + Flask/FastAPI companion API + Postgres via Compose
- ☐ Deploy on Minikube: Deployment, Service, Ingress manifests
- ☐ **Hero:** migrate to k3s on Oracle Cloud free tier, public URL

Skip CKA/CKAD/CKS (~\$395) and KCNA (~\$250) for now. A public k3s lab beats the badge.

PHASE 4

CI/CD Pipelines

Month 6

GitHub Actions deep; Jenkins only if a target role needs it

Skills

- ☐ GitHub Actions: workflows, jobs, matrix, secrets, environments, reusable workflows, OIDC to AWS
- ☐ Pipeline pattern: build → test → scan → package → deploy
- ☐ Deployment strategies: blue/green, canary, rolling, rollback
- ☐ Security scanning: Trivy, Checkov, Dependabot
- ☐ *Optional*: Jenkins, GitLab CI

Resources

Resource	Type	Cost
GitHub Actions docs + Skills labs	Official	Free
freeCodeCamp GitHub Actions course	YouTube	Free
CloudBees University Jenkins	Cert	Free
Trivy / Checkov docs	Docs	Free

Projects

- ☐ **Hero**: push → pytest → Docker build → Trivy scan → push registry → deploy to k3s
- ☐ Signed commits + branch protection
- ☐ OIDC from Actions to AWS (no long-lived keys)

PHASE 5

Infrastructure as Code

Month 7

Stop clicking

Skills

- ☐ Terraform: providers, resources, remote state (S3 + DynamoDB lock), modules, workspaces, for_each/count
- ☐ Terraform workflow: plan/apply in CI, module versioning
- ☐ Ansible: inventory, playbooks, roles, handlers, Jinja2, idempotency
- ☐ Concepts: declarative vs imperative, drift, immutable vs mutable

- ☐ Awareness: OpenTofu (open-source Terraform fork)

Resources

Resource	Type	Cost
HashiCorp Developer tutorials	Official	Free
Ansible official docs	Docs	Free
KodeKloud Terraform / Ansible	Course + badge	Verify still free
Stephane Maarek Terraform Associate	Udemy	~\$15 on sale

Projects

- ☐ **Hero:** Terraform all AWS infra (S3, Lambda, IAM, VPC), destroy/recreate in one command
- ☐ Ansible playbook for k3s node: install, SSH hardening, firewall
- ☐ Combined: Terraform provisions → Ansible configures → GitHub Actions deploys

Certification: HashiCorp Terraform Associate (003/004)

- ▶ Cost: **\$70.50 (~Rs. 23,100)**, no student discount
- ▶ ~57 questions, 60 min, online proctored, valid 2 years
- ☐ Pass exam

PHASE 6

Observability, Security & Advanced AWS

Months 8-9

Skills

- ☐ Prometheus: scrape configs, PromQL, exporters, alert rules
- ☐ Grafana: dashboards, data sources, variables, alerting
- ☐ Logging: CloudWatch Logs, structured logging, aggregation basics
- ☐ Cloud security: IAM policy JSON, Secrets Manager, KMS, VPC endpoints
- ☐ Advanced AWS: EKS overview, ECR, Route 53 routing, CloudFormation vs Terraform
- ☐ Cost: Cost Explorer, right-sizing, lifecycle rules, Reserved Instances basics

Resources

Resource	Type	Cost
Prometheus docs	Official	Free
Grafana Academy	Courses + badges	Free
PromLabs PromQL cheat sheet	Reference	Free
AWS Skill Builder Security path	Official	Free

Resource	Type	Cost
Stephane Maarek SAA-C03	Udemy	~\$15 on sale
Adrian Cantrill SAA-C03	Paid deep-dive	~\$40

Projects (finale)

- ☐ **Hero:** Prometheus + Grafana on k3s; dashboards for backup success rate, Lambda cold starts, API p50/p95/p99; Alertmanager to email
- ☐ README with architecture diagram (draw.io / Excalidraw)
- ☐ Blog post: "What I built in 9 months and what broke"
- ☐ Cloud cost optimisation write-up on the hero project's AWS bill

Certification: AWS Solutions Architect Associate (SAA-C03)

- ▶ Cost: \$150 (~Rs. 49,200). With 50% voucher from CLF-C02: **\$75 (~Rs. 24,600)**
- ▶ 65 questions, 130 min, 720/1000 to pass, valid 3 years

- ☐ Pass exam

Certification Timeline & Costs (verified Sep 2026)

Month	Certification	List	With discount	LKR
1-2	GitHub Foundations (GH-900)	\$99	Free (Student Pack)	Free
1-2	Harvard CS50P	Free	Free	Free
3-4	AWS Cloud Practitioner	\$100	\$50	Rs. 16,400
5	Docker/K8s badges (KodeKloud, if free)	Free	Free	Free
6	CloudBees Jenkins	Free	Free	Free
7	Terraform Associate	\$70.50	none	Rs. 23,100
8-9	AWS SAA-C03	\$150	\$75	Rs. 24,600

Realistic total: ~Rs. 64,100

Money-saving playbook

1. Apply to GitHub Student Developer Pack (free GH-900 voucher)
2. Register AWS Educate with university email (credits, labs, 50% voucher)
3. Buy Udemy courses only during sales (~\$15)
4. Watch re:Invent / AWS Summit for voucher drops
5. Pass CLF-C02 first; next-exam 50% voucher is automatic

Job Hunt (Month 8+)

Sri Lanka: WSO2, Sysco LABS, IFS, Virtusa, hSenid, 99x, Creative Software. Boards: devjobs.lk, LinkedIn (Cloud/DevOps, Sri Lanka)

Remote USD: Turing, We Work Remotely, Upwork (start early for reviews), RemoteOK, Working Nomads, Himalayas, LinkedIn Remote filter

Interview practice

- ▶ Interview before you feel ready; first 5 will be rough
- ▶ Pramp / interviewing.io mock interviews
- ▶ A "what went wrong" story for every project
- ▶ Know your salary floor; accepting ~30% less for the first role is realistic

Public Presence (from Month 1)

- ▶ GitHub: weekly commits, hero project public from day 1, one commit per finished project (no batching)
- ▶ Blog: one post per phase on dev.to or Hashnode
- ▶ LinkedIn: post each cert the day you earn it, project milestones, 2x/week
- ▶ X/Twitter: optional

Weekly Time Split (50 hrs)

Activity	Hours
Structured learning	18
Hands-on terminal / labs	15
Hero project work	12
Writing (blog + LinkedIn)	3
Review + plan (Sunday)	2

Progress Log

Date	Completed
Before Sep 2026	Linux Journey, Bandit 0-20, shellscrip.sh
02 Oct 2026	Addressbook CLI (Bash)

Living document. Update checkboxes and the log after every milestone. When prices, exam versions or tool licences change, update this file instead of trusting memory.